

ANALOGUE ASICs Made Simple



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Application-specific integrated circuits, or ASICs, typically conjure up the notion of massively-complex logic chips containing tens or hundreds of thousands (even millions) of transistors configured to solve a customer's unique set of problems. Unlike multifunction standard product ICs such as a microcontroller that can find its way into a wide variety of applications, ASICs are designed for one specific application and generally for one specific product or product family.

Today, most ASIC companies offer some degree of analogue functionality as a part of their services. In many cases, the analogue functions are mimicked with digital design techniques. In others, compromises to the analogue functionality must be made to facilitate the use of standard library cells that are designed to yield well in the fab processes developed for high-speed, high-density, low-power digital designs. Often, these chips are referred to as mixed-signal ASICs or as 'big D, little A' ASICs, meaning high digital content and minimal analogue content.

Analogue ASICs, or 'big A, little D' ASICs, play a critical role in our lives. Without these, none of the portable electronic devices we use in our daily lives would exist. Imagine a world without mobile phones, MP3 players and navigation systems. Building these with standard products would make these prohibitively expensive and physically impossible to carry in our purses or pockets.

Every automobile contains dozens of ASIC chips for everything from climate control to airbag deployment, suspension control to entertainment systems. ASICs also play important roles in applications for hospital medical equipment, e-meters, home appliances such as washers and dryers, scuba gear, hearing aids and much more.

The analogue ASIC market is huge. (See chart given in Table I from Summit Technology Consulting Inc.) Summit reports that almost 60 per cent of the nearly US\$ 42 billion of analogue ICs sold in 2014 were ASICs. Yet very few mixed-signal ASIC design houses fully understand the implications of custom analogue designs and its applicability to analogue-centric ASICs.

ASICs requiring high analogue content should be directed to those fabless semiconductor companies that specialise in analogue circuit designs rather than those who simply select analogue IP blocks from a library. Analogue ASIC companies have large staffs of competent, well-experienced analogue engineers with expertise in a wide range of analogue functions.

Large, standard product analogue IC companies rarely engage in analogue ASICs and, when they do, they set high bars regarding who can access this capability. They also impose high minimum order requirements. If you have a specific analogue need, you must rely on one of the many smaller companies that focus on low- to medium-volume analogue ASICs. Even if your volumes are high, these smaller companies will give you more direct attention and service because you are a more signifi-

TABLE I
ANALOGUE ASIC MARKET

Application-specific analogue ICs	2014 forecast (\$M)	Per cent of market
Consumer	1696	6%
Computer	2145	8%
Communications	13,002	50%
Automotive	6703	26%
Industrial/other	2715	10%
Total	26,261	94%
General purpose analogue ICs	2014 forecast (\$M)	Per cent of market
Amps and comparators	2729	16%
Interface	1940	11%
Voltage regulators and references	9831	58%
Data converters	2447	14%
Total	16,947	100%

Source: Summit Technology Consulting Inc.

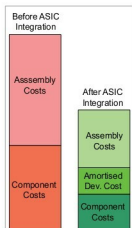


Fig. 1: Difference in the total system cost before and after ASIC integration